

AI-Based Early Detection of Cardiovascular Disease Using Medical Imaging

Abstract

Cardiovascular diseases are a major cause of morbidity and mortality worldwide. Early identification of cardiovascular abnormalities can improve clinical decision-making and patient outcomes. This proposal investigates the use of artificial intelligence and medical imaging to support earlier detection of cardiovascular disease.

Research Problem

Current diagnostic workflows may require significant specialist time and can be affected by variations in image interpretation. There is a need for reliable computational methods that can assist clinicians in identifying potentially important patterns in medical images.

Objectives

1. Develop a machine learning model for identifying cardiovascular abnormalities from medical imaging data.
2. Evaluate the model using accuracy, precision, recall, and F1-score.
3. Compare the proposed approach with selected baseline machine learning methods.
4. Assess the potential usefulness of the system as a clinical decision-support tool.

Methodology

The study will use an anonymized medical imaging dataset. Images will be preprocessed and divided into training, validation, and test sets. A deep learning model will be trained on the training data and evaluated on unseen test data. Model performance will be measured using standard classification metrics.

Expected Outcomes

The expected outcome is a validated prototype capable of assisting with early identification of cardiovascular abnormalities. The study will also identify limitations, possible sources of error, and directions for future clinical validation.

Keywords

Artificial intelligence, cardiovascular disease, medical imaging, machine learning, deep learning, clinical decision support.